DO NOW — RECALL

Network types, architecture and hardware

Name: _____ Class: _____ Date: _____

1. What does an OS device driver do?

2. What does a router connect?

3. Define a client-server relationship.

4. Binary search on sorted indexes 0–14 for 13: show midpoint index sequence.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Offers an interface to hardware.

2. Model answer

Different IP networks.

3. Model answer

Client requests service from a dedicated server.

4. Model answer

mid7, mid11, mid13; target found at index13.

Vocabulary — know these terms

LAN — Local Area Network.

WAN — Wide Area Network.

switch — Forwards frames inside local network.

router — Forwards packets between IP networks.

access point — Provides wireless connection to LAN.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check the answer

switch connects local devices

client-server centralises authentication/files.

IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Two classrooms use Wi-Fi but need access to a central school file server. Identify useful hardware/architectur

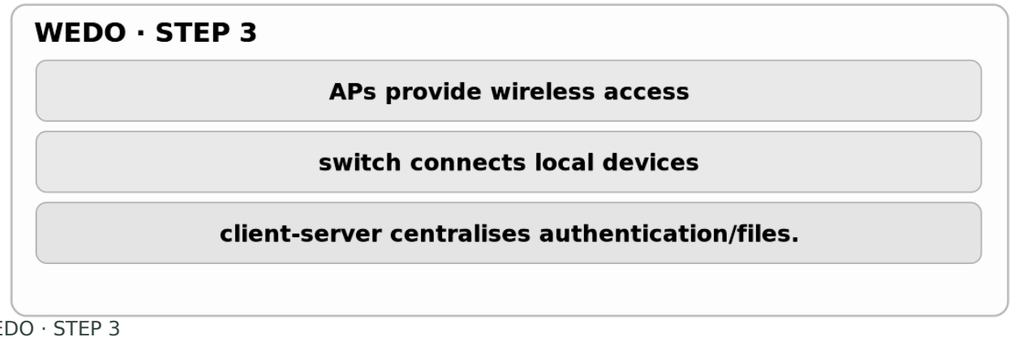
APs provide wireless access



WEDO · STEP 2

Two classrooms use Wi-Fi but need access to a central school file server. Identify useful hardware/architectur

APs provide wireless access



WEDO · STEP 3

NEW LEARNING

Read, annotate and explain

Understand

A LAN covers a local area such as a school building; a WAN connects networks across wider geography.

Trace

Client-server uses dedicated service systems; peer-to-peer allows peers to exchange roles and resources, with different management implications.

Explain

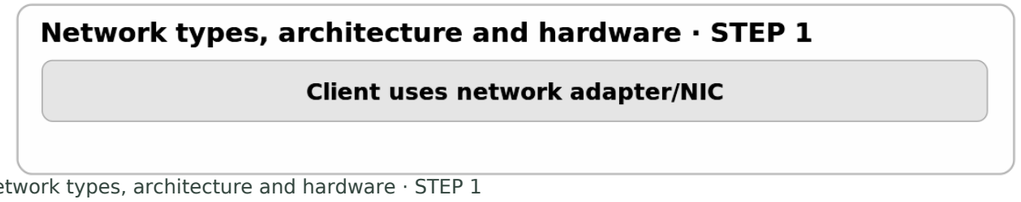
NICs connect devices, switches forward local Ethernet frames, wireless APs bridge Wi-Fi, and routers forward IP packets between networks.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



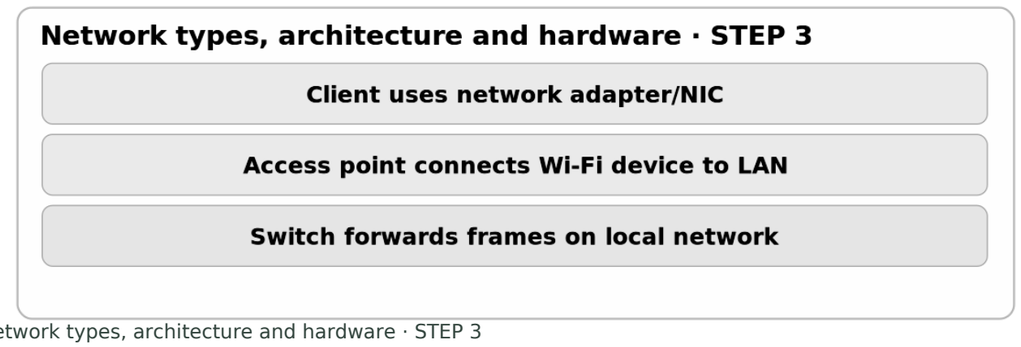
Network types, architecture and hardware · STEP 1

New Learning



Network types, architecture and hardware · STEP 2

New Learning



Network types, architecture and hardware · STEP 3

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A network switch and router are always interchangeable because they both move packets.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

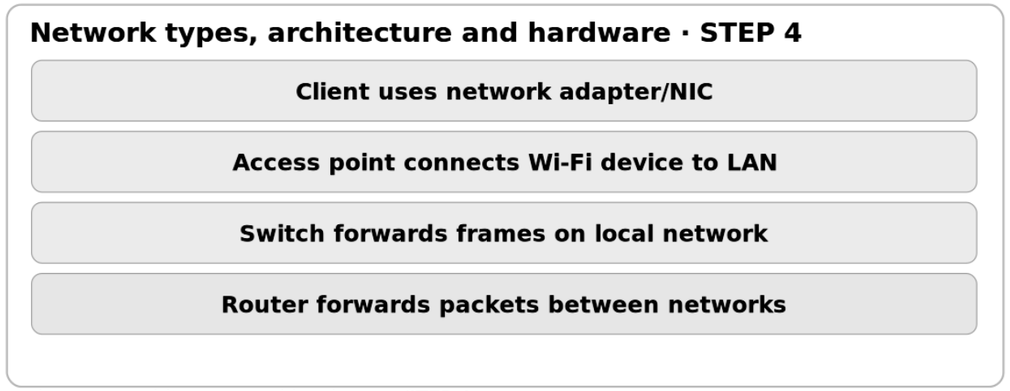
Question 3

Compare client-server and peer-to-peer for school file sharing, giving strengths and limits. [6]

New Learning · original lesson diagrams

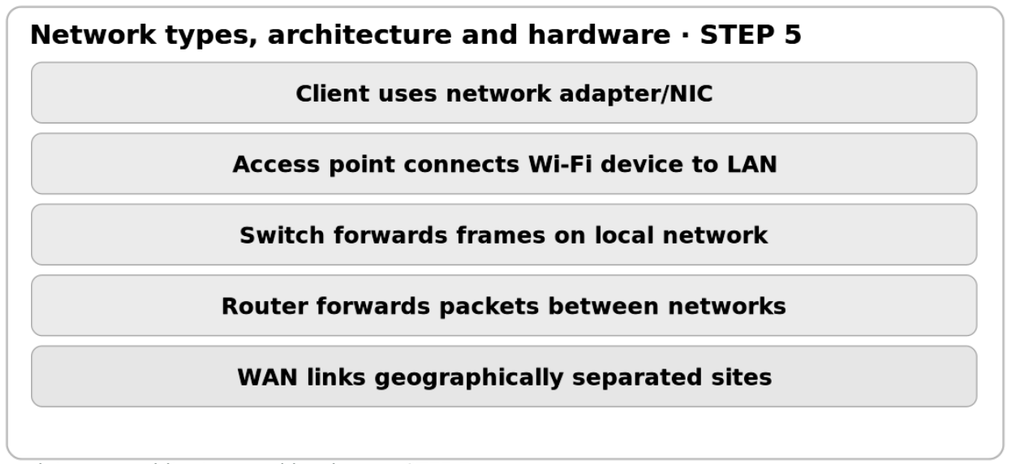
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Network types, architecture and hardware · STEP 4

New Learning



Network types, architecture and hardware · STEP 5

I DO — WORKED EXAMPLE

Follow the worked method

Trace school laptop reaching an external site through a wired school network.

Predict

Switch moves frames within LAN

Check the answer

router handles packets destined for another network.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define LAN, WAN, switch and router. [4]

Question 2

Describe a laptop's network path via wireless access point, switch and router to the Internet. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict APs provide wireless access

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace school laptop reaching an external site through a wired school network.

Switch moves frames within LAN



IDO · STEP 1

Trace school laptop reaching an external site through a wired school network.

Switch moves frames within LAN



IDO · STEP 2

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Two classrooms use Wi-Fi but need access to a central school file server. Identify useful hardware/architectur

APs provide wireless access



WEDO · STEP 1